

Unit 4

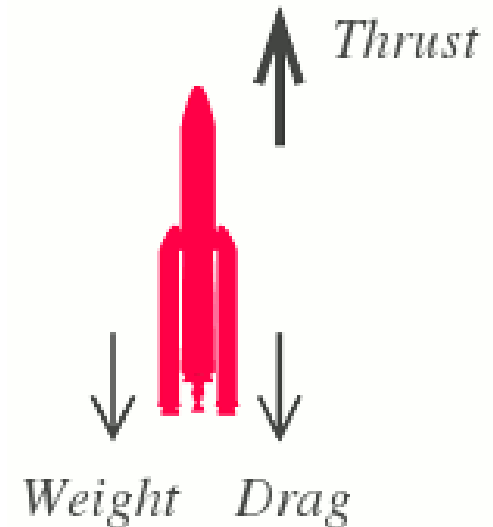
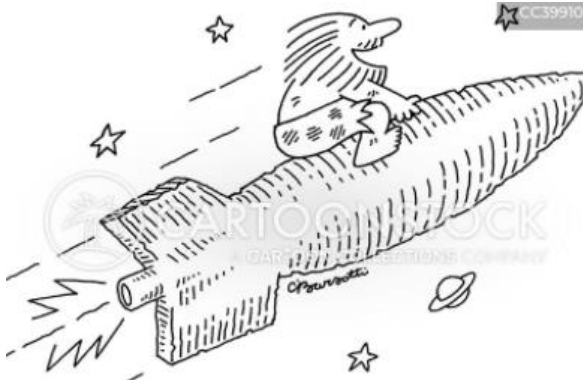
Rocket Performance Parameters



ROCKETS



- Rocket- ***Jet powered vehicle***
- Deliver **Payloads**
- Similar to Trucks, Aeroplane, Ships etc
- Delivery- ***earth surface to space***
- Propelled by ***Thrust***
- Thrust-Primary force- **NEWTONS Laws**



Agency 1

Rocket A



**SOLID PROPELLANT
ROCKET**

Agency 2

Rocket A



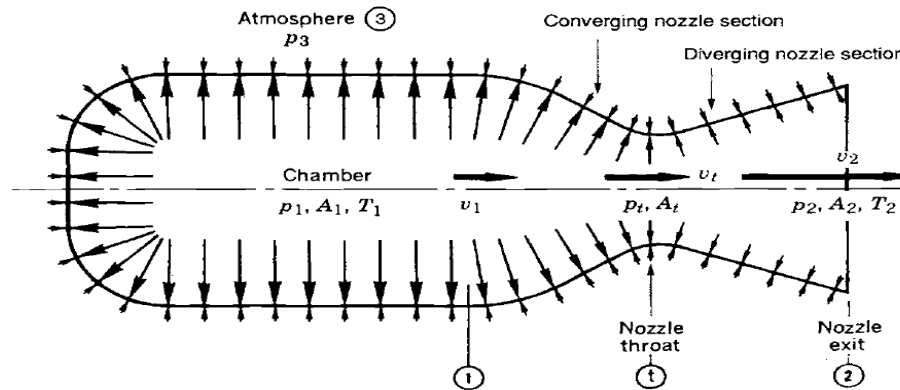
**LIQUID PROPELLANT
ROCKET**

Performance Parameters

- *Thrust (T or F)*
- *Effective Exhaust velocity (V_j or c)*
- *Specific Impulse (I_{sp})*
- *Total Impulse (I_T)*
- *Specific Propellant Consumption (Spc)*
- *Thrust Coefficient (C_F)*
- *Impulse-to-weight Ratio (IWR)*
- *Thrust-to-weight Ratio (T/w)*
- *Characteristic Jet Velocity (C^*)*
- *Mass Ratio (MR)*
- *Propellant Mass Fraction (ζ)*

Thrust (T or F)

- **Combustion - Thrust**
- Thrust – Two Factors Contribute - **Momentum & Pressure**



$$F_{Total} = F_{Momentum} + F_{Pressure}$$

$$F_{Momentum} = \frac{dm}{dt} v_2$$

$$F_{Momentum} = \dot{m} v_2$$

$$F_{Pressure} = (p_2 - p_3) A_2$$

$$F = \dot{m} v_2 + (p_2 - p_3) A_2$$

Based on the Operating condition of the Nozzle

Case 1 – $p_2 > p_3$: Under expanded Condition - Positive Thrust

$$F = \dot{m}v_2 + (p_2 - p_3)A_2$$



For example consider a rocket moving at an altitude of 10km from the launch pad. The following are the conditions of the rocket:

Nozzle Throat Area A_t : 0.036m

Nozzle Exit Area A_e : 0.0807m

Nozzle Area Ratio AR : 4.80

Thrust Chamber Pressure p_c : 24.68 Bar

Thrust Chamber Temp T_c : 1926°C

At $h_0=10\text{km}$, p_a : 0.245 Bar and T_a : -28°C

For given conditions and also for the mass, density and other operating conditions, we get the exit pressure $p_e = 0.55$ Bar at the exit plane of the nozzle.

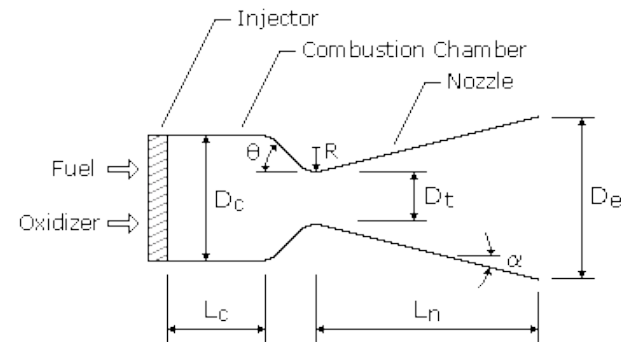


Figure 1.4

$$F = \dot{m}v_2 + (0.55 - 0.245) A_2$$

Case 2 – $p_2 = p_3$: Optimally expanded Condition - Zero Thrust

$$F = \dot{m}v_2 + (0)A_2$$

$$F = \dot{m}v_2$$

From previous example,

$$F = \dot{m}v_2 + (0.245 - 0.245)A_2$$



Case 3 – $p_2 < p_3$: Over expanded Condition- Negative Thrust (Very Low & Undesirable Thrust)

$$F = \dot{m}v_2 + (0.102 - 0.245)A_2$$



- **Case 4 – $p_3=0$: In Vacuum-Maximum Thrust (Because $p_3=0$)**

$$F = \dot{m}v_2 + p_2A_2$$

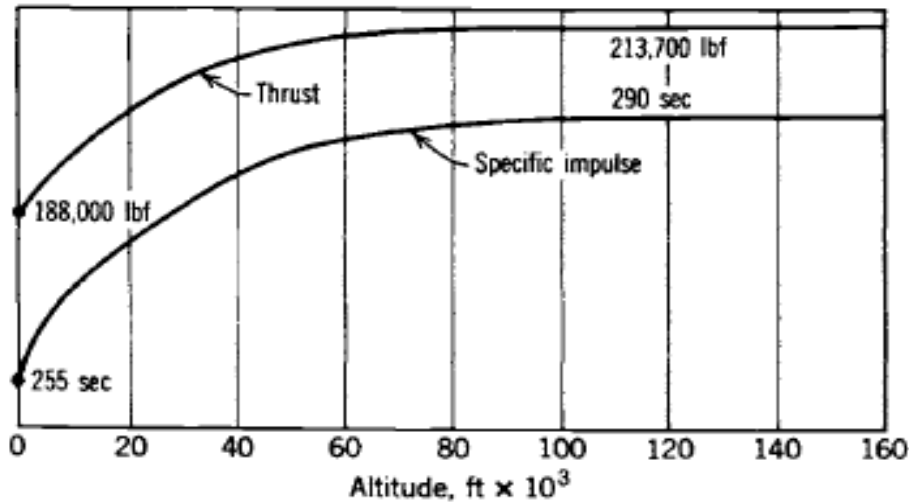


Figure 2: Thrust and Specific Impulse Increases with altitude

Rocket Parameters

Rocket	Delta II Space Launch booster RS-27	Booster stage for H-II launcher LE-7 (Japan)
Thrust Sea Level	732.622 kN	846.941 kN
Thrust Altitude	920.781 kN	1078.693 kN
Specific Impulse Sea Level	257	349.9
Specific Impulse Altitude	294	445.6
Thrust chamber propellant flow	290.299 kg/s	157.35 kg/s
Characteristic exhaust velocity, c^*	1688.592 m/s	1705.29504 m/s



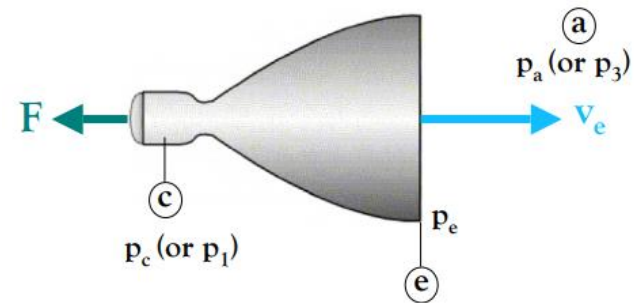
Delta II Space Launch booster RS-27



Booster stage for H-II launcher LE-7 (Japan)

EFFECTIVE EXHAUST VELOCITY (c or v_j)

- Nozzle-Pressure to Velocity
- Maximum Velocity - Nozzle Exit Plane Specifically - **Centreline**
- ***Flow Losses, Viscous Effect-Skin Friction Losses, Flow Separation, Divergence Losses, Two-Phase Flows, Shocks etc***
- Velocity Distribution - Not Uniform-Entire Cross Section
- So is the THRUST
- Doesn't Represent entire **Thrust Magnitude**
- For convenience uniform axial velocity c is assumed- exhaust velocity .
- **Effective Exhaust Velocity v_j - Average Equivalent Velocity** at which propellant is ejected from the vehicle
- At the design condition where $p_e = p_a$



EFFECTIVE EXHAUST VELOCITY (c or v_j) Contd...

$$F = \dot{m}v_e + (p_e - p_a)A_e$$

$$p_e = p_a \quad v_e = v_j$$

$$F = \dot{m}v_j$$

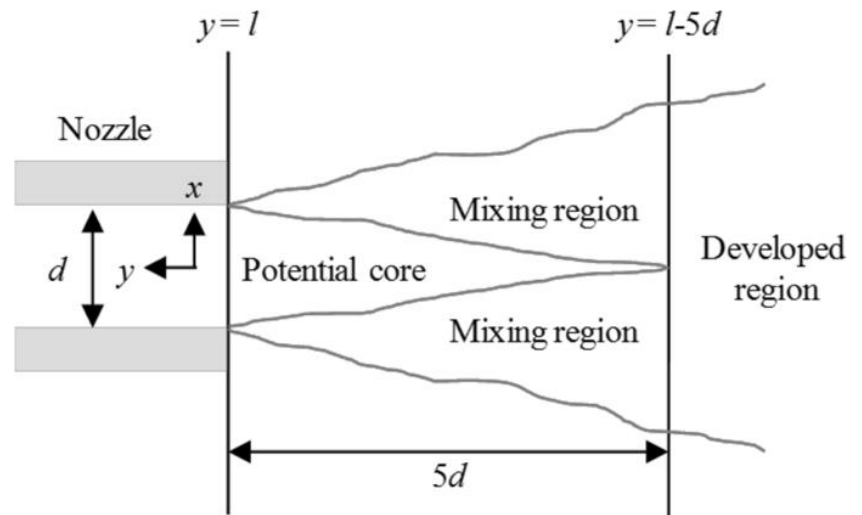
$$v_j = \frac{F}{\dot{m}}$$

$$v_j = \frac{F}{\frac{\dot{w}}{g}}$$

$$v_j = \frac{F * g}{\dot{w}}$$

$$I_{sp} = \frac{F}{\dot{w}}$$

$$v_j = I_{sp} * g$$



SPECIFIC IMPULSE (I_{sp})

- Important figure of merit
- Ratio of thrust per (F) unit weight flow rate ($\dot{w}_p$) of propellant
- Similar to **MILEAGE** in Automobiles

$$I_{SP} = \frac{F}{\dot{W}_p} \quad \text{Seconds}$$

$$I_{SP} = \frac{\frac{\dot{W}_p}{g} * v_j}{\dot{W}_p}$$

$$I_{SP} = \frac{v_j}{g} \text{Seconds}$$

SPECIFIC IMPULSE (I_{sp}) Contd....

- Also defined as ratio of total impulse (I_t) to weight flow rate of propellants (w_p)

$$I_s = \frac{\int_0^t F dt}{g_0 \int \dot{m} dt}$$

OBJECT	ACCELERATION DUE TO GRAVITY	GRAVITY
Earth	9.8 m/s ² or 32 ft/s ²	1 G
the Moon	1.6 m/s ² or 5.3 ft/s ²	.16 G
Mars	3.7 m/s ² or 12.2 ft/s ²	.38 G
Venus	8.87 m/s ² or 29 ft/s ²	0.9 G
Jupiter	24.5 m/s ² or 80 ft/s ²	2.54
the Sun	275 m/s ² or 896 ft/s ²	28 G

TOTAL IMPULSE (I_t)

- Total impulse is the thrust force integrated over the burn time t .

$$I_t = \int_0^t F * dt$$

- For constant thrust, that does not vary with time,

$$I_t = F \int_0^t dt$$

$$I_t = Ft$$

$$I_s = \frac{F}{\dot{w}_p}$$

$$F = I_{sp} * \dot{w}_p$$

$$I_t = \int_0^t F * dt$$

$$I_t = \int_0^t I_{sp} \dot{w}_p dt$$

TOTAL IMPULSE (I_t)

- I_{sp} is a constant value for a given propellant. So

$$I_t = I_{sp} \int_0^t \dot{w}_p dt$$

$$I_t = I_{sp} * w_p$$

- Different Total Impulse can be achieved by varying I_{sp} and t

$$I_t = \int_0^t I_{sp} \dot{w}_p dt$$

SPECIFIC PROPELLANT CONSUMPTION (SPC)

- Defined as the weight flow rate of the propellant ($\dot{w}_p$) required to produce 1Newton of thrust (F)
- SPC is just the inverse of specific impulse

$$SPC = \frac{\dot{w}_p}{F}$$

$$SPC = \frac{1}{\frac{F}{\dot{w}_p}}$$

$$SPC = \frac{1}{I_{sp}}$$

THRUST COEFFICIENT (C_F)

- Propellants- High Thrust & Short Burn Time
- Optimum Performance - **Nozzle Design** - Paramount Importance
- Governed by **Thrust Coefficient C_F**
- **Thrust Coefficient C_F** . Thrust (F) per unit chamber pressure (p_c) and throat area A^*

$$C_F = \frac{F}{p_c A^*}$$

THRUST COEFFICIENT (C_F) Contd...

- Signifies – Degree of Thrust Amplification
- Conversion of High Pressure to Low Pressure
- Thrust coefficient can be expressed in terms of characteristic velocity c^* and specific impulse I_{sp} . Its given as

$$C_F = \frac{I_{sp}}{c^*}$$

- Well designed nozzle- C_F = Under 1.5
- Thrust coefficient will be MAXIMUM when $p_e = p_a$

IMPULSE-TO-WEIGHT RATIO (IWR)

- Ratio of total impulse (I_T) of the rocket to the total weight (w) of the rocket at take-off

$$IWR = \frac{I_t}{wt}$$

$$I_t = I_s * w_p$$

$$IWR = \frac{I_s * w_p}{wt}$$

$$IWR = Isp * \left(\frac{w_p}{wt} \right)$$

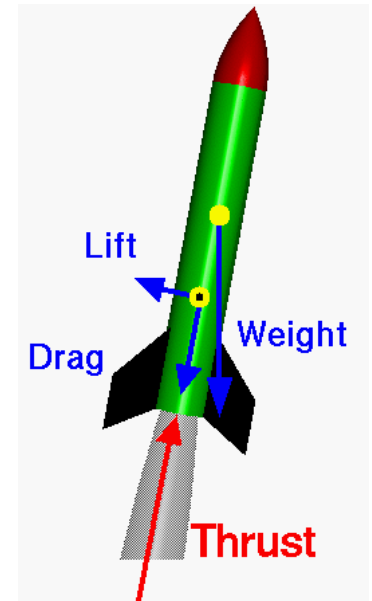
$$\left(\frac{W_p}{W_t} \right) = \zeta$$

- ζ = Propellant Mass Fraction

THRUST-TO-WEIGHT RATIO (F/w)

- Ratio of thrust (F) generated to the total weight (w) of the rocket at take-off

$$TWR = \frac{F}{w}$$



- Signifies **ACCELERATION** of the vehicle
- Acceleration varies with altitude
- Maximum Acceleration- ***Just Before Burn Out***

Thrust-to-Weight Ratio Examples

Rocket	Mass kg	Thrust (Vacuum) kN	TWR
RD-0410 Nuclear Rocket Engine	2000	35.2	1.8
Rocketdyne RS-25	3177	2278	73.1
Merlin 1D	467	825	180.1

CHARACTERISTIC JET VELOCITY c^*

- Ratio of effective jet velocity v_j to the thrust coefficient C_F

$$c^* = \frac{V_j}{C_F}$$

- Multiply & Divide by g

$$c^* = \frac{V_j}{g} * \frac{g}{C_F}$$

$$c^* = \frac{I_{sp} * g}{C_F}$$

$$c^* = \frac{g}{C_F * SPC}$$

MASS RATIO (MR)

- Ratio of the final mass m_f of the rocket after complete burnout to the total mass m_i of the rocket at take-off.

$$MR = \frac{m_f}{m_i}$$

- m_f = mass of the rocket after complete burnout (after all the propellants have been consumed)
(Payload + Vehicle Mass)
- m_i = mass of the rocket at take-off **(Payload + Vehicle Mass + Propellant)**

- Mass Ratio Measures the Efficiency of the Rocket**

- Delta II / 7920**

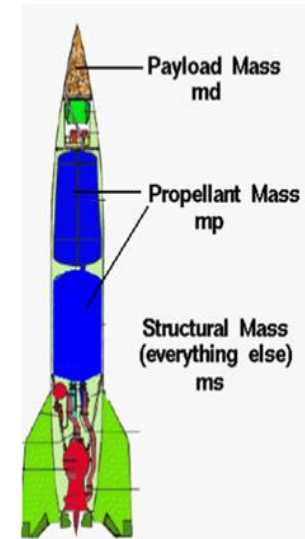
Propellant, strap-ons: Solid

Propellant, 1-stage: LOX/Kerosene

Propellant, 2-stage: Nitrogen tetroxide/Aerozine 50

Launch mass: 227,200 kg + payload 5039 kg

Propellant mass: 207,900 kg



- A Delta 7920 can deliver 5,039 kg to low earth orbit from Cape Canaveral, but it can only deliver 3,175 kg to a Sun-synchronous orbit from Vandenberg Air Force Base.

- **Delta II / 7920**

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Propellant, 1-stage: LOX/Kerosene

Propellant, 2-stage: Nitrogen tetroxide/Aerozine 50

Launch mass: 227,200 kg + payload 5039 kg

Propellant mass: 207,900 kg

- Thus, in the first case the mass ratio is: $MR = \frac{m_f}{m_i}$ $MR = 0.5276$

$$MR = \frac{m_f}{m_i} \quad MR = 0.5256$$

PROPELLANT MASS FRACTION ζ

- ζ indicates the fraction of propellant mass m_p in an initial mass m_0
- Defined as the ratio of mass of propellant m_p at take-off to the total mass of the rocket at take-off m_0 .

$$\zeta = \frac{m_p}{m_0}$$

$$m_0 = m_f + m_p \quad m_p = m_0 - m_f$$

$$\zeta = \frac{m_0 - m_f}{m_0}$$

$$\zeta = 1 - \frac{m_f}{m_0}$$

$$\zeta = 1 - MR$$

- **Delta II / 7920**

Propellant, strap-ons: Solid

Propellant, 1-stage: LOX/Kerosene

Propellant, 2-stage: Nitrogen tetroxide/Aerozine 50

Launch mass: 227,200 kg + payload 5039 kg

Propellant mass: 207,900 kg

$$\zeta = 1 - \frac{m_f}{m_o}$$

$$\zeta = 1 - \frac{(227200 + 5039)}{(227200 + 5039 + 207900)}$$

$$\zeta = 1 - 0.5276$$

$$\zeta = 0.4723$$



Numericals

A spacecraft's engine ejects mass at a rate of 30 kg/s with an exhaust velocity of 3,100 m/s. The pressure at the nozzle exit is 5 kPa and the exit area is 0.7 m². What is the thrust of the engine in a vacuum?

Parameters Given:

- $\dot{m} = 30 \text{ kg/s}$
- $V_e = 3,100 \text{ m/s}$
- $A_e = 0.7 \text{ m}^2$
- $P_e = 5 \text{ kPa} = 5,000 \text{ N/m}^2$
- $P_a = 0$

$$F = \dot{m} \cdot V_e + (P_e - P_a) \times A_e$$

$$F = 30 \times 3,100 + (5,000 - 0) \times 0.7$$

$$F = 96,500 \text{ N - Thrust in Vacuum}$$

A rocket engine produces a thrust of 1,000 kN at sea level with a propellant flow rate of 400 kg/s. Calculate the specific impulse.

SOLUTION,

Given: $F = 1,000,000 \text{ N}$

$\dot{m} = 400 \text{ kg/s}$

$$I_{sp} = F / (\dot{m} \times g)$$

$$I_{sp} = 1,000,000 / (400 \times 9.80665)$$

$$I_{sp} = 255 \text{ s (sea level)}$$

A rocket projectile has the following characteristics:

Initial mass: 200 kg

Mass after rocket operation : 130 kg

Payload, nonpropulsive structure, etc. : 110 kg

Rocket operating duration : 3.0 sec

Average specific impulse of propellant : 240 sec

- Determine the vehicle's mass ratio, propellant mass fraction, propellant flow rate, thrust, thrust-to-weight ratio, acceleration of vehicle, effective exhaust velocity, total impulse, and the impulse-to-weight ratio.

SOLUTION

- *Mass Ratio of the Vehicle System*

$$MR = \frac{m_f}{m_i} \qquad MR = \frac{130}{200} = 0.65$$

- *Mass Ratio of the Propellant System*

$$MR = \frac{m_f}{m_i} \qquad MR = \frac{130 - 110}{200 - 110} = 0.22$$

- *Note that empty and initial masses of the propulsion system are 20 and 90kg*

- **Propellant mass fraction ζ**

$$\zeta = \frac{m_p}{m_o}$$

$$\zeta = \frac{m_o - m_f}{m_o}$$

$$\zeta = \frac{90 - 20}{90} = 0.778$$

- **Propellant mass is 200-130kg=70kg**
- **Propellant Mass flow rate $\dot{m}=70/3=23.3\text{kg/sec}$**
- **Thrust of the rocket**

$$F = I_{sp} * \dot{w}$$

$$F = 240 * 23.3 * 9.81$$

$$F = 54857 \text{ Newtons}$$

- ***Thrust-to-weight ratio of the vehicle is***

$$(TWR)_{Initial} = \frac{F}{\dot{W}_o}$$

$$(TWR)_{Initial} = \frac{54857}{200 * 9.81}$$

$$(TWR)_{Initial} = 28$$

$$(TWR)_{Final} = \frac{F}{\dot{W}_o}$$

$$(TWR)_{Final} = \frac{54857}{130 * 9.81}$$

$$(TWR)_{Final} = 43$$

- ***The effective exhaust velocity c^* or v_j***

$$V_j = I_{sp} * g_0$$

$$V_j = 240 * 9.81 = 2354m / s$$

- **Total impulse I_t**

$$I_t = I_{sp} * wp$$

$$I_t = 240 * 70 * 9.81$$

$$I_t = 164808 \text{ Newtons}$$

- **Impulse-to-weight ratio IWR**

$$IWR = \frac{I_t}{w_0}$$

$$IWR = \frac{164808}{(200 - 110) * 9.81}$$

$$IWR = 187$$

A spacecraft with an initial mass of 30000kg is just launched from the launch pad. 25.5kg of exhaust gas is being ejected out from the nozzle at an exhaust velocity of 2700m/s. For this condition calculate what will be the change in velocity if the spacecraft burns its engine for 180 seconds

SOLUTION

Given: $m_0 = 30,000 \text{ kg}$
 $\dot{m} = 25.5 \text{ kg/s}$
 $V_e = 2700 \text{ m/s}$
 $t = 180 \text{ s}$

$$\Delta V = v_e \ln \frac{m_0}{m_1}$$

$$\Delta V = ?$$

$$\Delta V = v_e \ln \frac{m_0}{(m_0 - \dot{m} * t)}$$

$$\Delta V = 2700 \ln \frac{30000}{(30000 - (25.5 * 180))}$$

$$\Delta V = 448.3 \text{ m/s}$$

The definition arises naturally from [Tsiolkovsky's rocket equation](#):

$$\Delta v = v_e \ln \frac{m_0}{m_1}$$

where

- Δv is the desired change in the rocket's velocity
- v_e is the effective exhaust velocity (see [specific impulse](#))
- m_0 is the initial mass (rocket plus contents plus propellant)
- m_1 is the final mass (rocket plus contents)

This equation can be rewritten in the following equivalent form:

$$\frac{m_0}{m_1} = e^{\Delta v / v_e}$$

A spacecraft's dry mass is 75,000 kg and the effective exhaust gas velocity of its main engine is 3,100 m/s. How much propellant must be carried if the propulsion system is to produce a total Δv of 700 m/s?

SOLUTION

Given: $m_f = 75,000 \text{ kg}$

$V_e = 3,100 \text{ m/s}$

$\Delta v = 700 \text{ m/s}$

$$\Delta V = v_e \ln \frac{m_0}{m_1}$$

Dry Mass= After all the propellant is consumed

$$\frac{m_0}{m_1} = e^{\frac{\Delta v}{v_e}}$$

$$m_0 = m_1 \times e^{(\Delta v / v_e)}$$

$$m_0 = 75,000 \times e^{(700 / 3,100)}$$

$$m_0 = 94,000 \text{ kg}$$

Propellant mass,

$$\begin{aligned} m_p &= m_0 - m_1 \\ m_p &= 94,000 - 75,000 \\ m_p &= 19,000 \text{ kg} \end{aligned}$$

A 5,000 kg spacecraft is in Earth orbit traveling at a velocity of 7,790 m/s. Its engine is burned to accelerate it to a velocity of 12,000 m/s placing it on an escape trajectory. The engine expels mass at a rate of 10 kg/s and an effective velocity of 3,000 m/s. Calculate the duration of the burn.

SOLUTION,

$$\text{Given: } m_0 = 5,000 \text{ kg}$$

$$\dot{m} = 10 \text{ kg/s}$$

$$v_e = 3,000 \text{ m/s}$$

$$\Delta v = 12000 - 7790 \text{ m/s}$$

$$\Delta v = 4210 \text{ m/s}$$

$$\Delta V = v_e \ln \frac{m_0}{m_1}$$

$$\Delta V = v_e \ln \frac{m_0}{(m_0 - \dot{m} * t)}$$

$$\Delta V = v_e \ln \frac{m_0}{(m_0 - t)}$$

$$\frac{\Delta V}{v_e} = \ln \frac{m_0}{(m_0 - mf * t)}$$

$$e^{\frac{\Delta V}{v_e}} = \frac{m_0}{(m_0 - mf * t)}$$

$$(m_0 - mf * t) e^{\frac{\Delta V}{v_e}} = m_0$$

$$(m_0 - mf * t) = \frac{m_0}{e^{\frac{\Delta V}{v_e}}}$$

$$t = \left(\frac{m_0}{\dot{m}} \right) \left(1 - \frac{1}{e^{\frac{\Delta V}{v_e}}} \right)$$

$$t = 500 \times [1 - 1 / e(4,210 / 3,000)]$$

$$\mathbf{t = 377.1 \text{ s}}$$

A solid rocket motor burns along the face of a central cylindrical channel 10 meters long and 1 meter in diameter. The propellant has a burn rate coefficient of 5.5, a pressure exponent of 0.4, and a density of 1.70 g/ml. Calculate the burn rate and the product generation rate when the chamber pressure is 5.0 MPa.

SOLUTION,

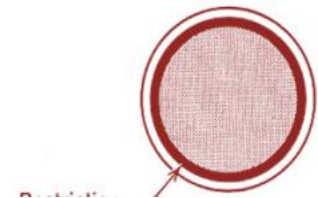
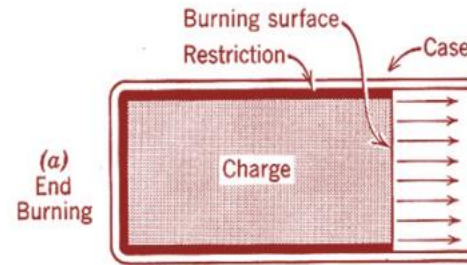
Given: $a = 5.5$

$n = 0.4$

$P_c = 5.0 \text{ MPa}$

$\rho = 1.70 \text{ g/ml}$

$A_b = \pi \times 1 \times 10 = 31.416 \text{ m}^2$



$$r = a \times P_c^n$$

$$r = 5.5 \times 5.0^{0.4} = 10.47 \text{ mm/s}$$

$$\dot{m} = \rho \times A_b \times r$$

$$\dot{m} = 1.70 \times 31.416 \times 10.47 = 559 \text{ kg/s}$$

A two-stage rocket has the following masses: 1st-stage propellant mass 120,000 kg, 1st-stage dry mass 9,000 kg, 2nd-stage propellant mass 30,000 kg, 2nd-stage dry mass 3,000 kg, and payload mass 3,000 kg. The specific impulses of the 1st and 2nd stages are 260 s and 320 s respectively. Calculate the rocket's total V.

SOLUTION,

Given:

$$M_{o1} = 120,000 + 9,000 + 30,000 + 3,000 + 3,000 = 165,000 \text{ kg} = \text{TOTAL MASS AT TAKE-OFF}$$

$$M_{f1} = 9,000 + 30,000 + 3,000 + 3,000 = 45,000 \text{ kg}$$

$$I_{sp1} = 260 \text{ s}$$

$$M_{o2} = 30,000 + 3,000 + 3,000 = 36,000 \text{ kg}$$

$$M_{f2} = 3,000 + 3,000 = 6,000 \text{ kg}$$

$$I_{sp2} = 320 \text{ s}$$

$$C1 = I_{sp1} * g$$

$$C1 = 260 \times 9.80665 = 2,550 \text{ m/s}$$

$$C2 = I_{sp2} * g$$

$$C2 = 320 \times 9.80665 = 3,138 \text{ m/s}$$

$$\mathbf{V_1 = C_1 \times \ln [Mo_1 / Mf_1]}$$

$$V_1 = 2,550 \times \ln [165,000 / 45,000]$$

$$V_1 = 3,313 \text{ m/s}$$

$$\mathbf{V_2 = C_2 \times \ln [Mo_2 / Mf_2]}$$

$$V_2 = 3,138 \times \ln [36,000 / 6,000]$$

$$V_2 = 5,623 \text{ m/s}$$

$$\mathbf{V_{Total} = V_1 + V_2}$$

$$V_{Total} = 3,313 + 5,623$$

$$V_{Total} = 8,936 \text{ m/s}$$

Calculate the velocity of an artificial satellite orbiting the Earth in a circular orbit at an altitude of 200 km above the Earth's surface.

SOLUTION,

From Basics Constants,

Radius of Earth = 6,378.14 km

GM of Earth = $3.986005 \times 10^{14} \text{ m}^3/\text{s}^2$

Given: $r = (6,378.14 + 200) \times 1,000 = 6,578,140 \text{ m}$

$$v = \text{SQRT}[GM / r]$$

$$v = \text{SQRT}[3.986005 \times 10^{14} / 6,578,140]$$

$$v = 7,784 \text{ m/s}$$

Calculate the escape velocity of a spacecraft launched from an Earth orbit with an altitude of 200 km.

SOLUTION,

$$\text{Given: } r = (6,378.14 + 200) \times 1,000 = 6,578,140 \text{ m}$$

$$V_{\text{esc}} = \text{SQRT}[2 \times GM / r]$$

$$V_{\text{esc}} = \text{SQRT}[2 \times 3.986005 \times 10^{14} / 6,578,140]$$

$$V_{\text{esc}} = 11,009 \text{ m/s}$$